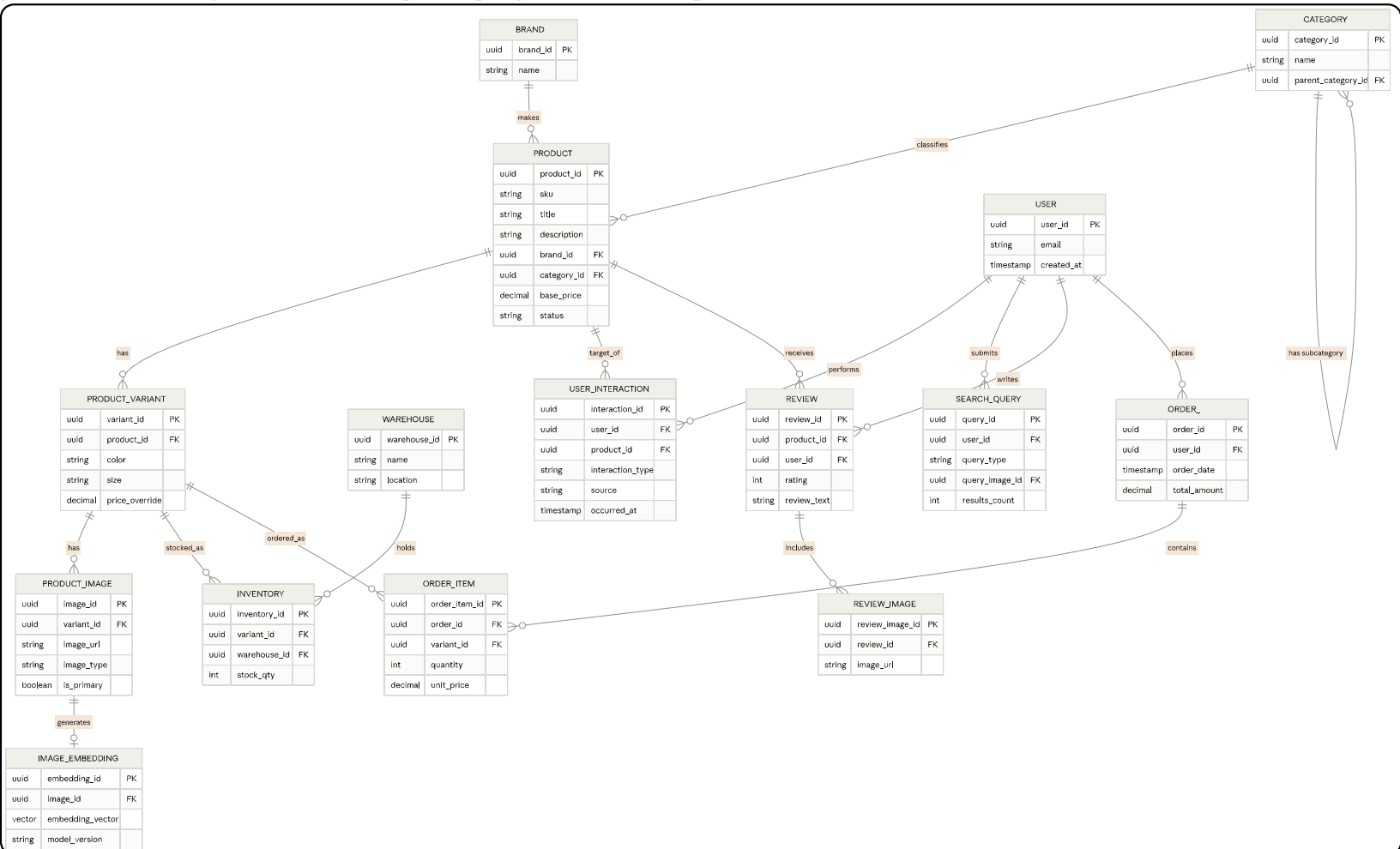


# Entity-Relationship Diagram — Visual Product Search Data Model

This diagram renders natively on GitHub (mermaid.js). It covers every entity needed to support visual product search: product catalog, imagery + embeddings, user behavior, and transactions.



## How to read this

**Catalog spine:** **BRAND** and **CATEGORY** classify **PRODUCT**; each **PRODUCT** has one or more **PRODUCT\_VARIANT** (color/size combinations), and each variant has its own images and stock.

**Visual search core:** **PRODUCT\_IMAGE** → **IMAGE\_EMBEDDING** is the pair that actually powers visual search — every catalog image gets a vector embedding used for similarity lookup. **SEARCH\_QUERY** can itself reference an uploaded image (**query\_image\_id**), which is embedded the same way and compared against **IMAGE\_EMBEDDING**.

**Behavioral loop:** **USER\_INTERACTION** and **SEARCH\_QUERY** capture what users do and search for; **REVIEW** (with its own **REVIEW\_IMAGE**s) is a second, real-world source of product imagery distinct from staged catalog photos.

**Transactional spine:** **ORDER\_** → **ORDER\_ITEM** → **PRODUCT\_VARIANT** ties visual search usage back to actual revenue, and **INVENTORY** (per variant, per **WAREHOUSE**) supports "in stock near me" filtering on search results.

See **data\_dictionary.md** for every field's type and purpose, and **modeling\_rationale.md** for why this is structured as normalized core + a denormalized search layer rather than one or the other.